# Recon Drone: MCM Edutech Asia

Weights & Biases training report

Project marlborough-college-malaysia/Edutech

## 1 Overview

| Run | W&B id | State | Params | GFLOPs | Speed (ms) | Classes | Dataset | Epochs logged / set | mAP50 | mAP50-95 | Precision | Recall | Runtime |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| likely-fog-13 | dsku6opn | finished | 3.0 M | 8.2 | 1.2 | 12 | 12-class marine-debris set | 3 / 3 | 0.578 | 0.413 | 0.614 | 0.542 | 3 min |
| ethereal-sound-16 | qv9gydoe | crashed | 56.9 M | 195.7 | 15.0 | 40 | 40-class set | 3 / 5 | 0.418 | 0.315 | 0.449 | 0.447 | 108 min |
| sleek-dream-43 | bqhlcnnf | finished | 25.3 M | 87.3 | 1.3 | 8 | 8-class set, Roboflow marine-debris-vb2qm | 5 / 100 | 0.324 | 0.218 | 0.529 | 0.330 | 44 min |
| apricot-elevator-45 | cxareo1p | finished | 25.3 M | 87.3 | 1.2 | 8 | 8-class set, as sleek-dream-43 | 1 / 1 | 0.063 | 0.033 | 0.240 | 0.129 | 12 min |

## 2 Detection metrics

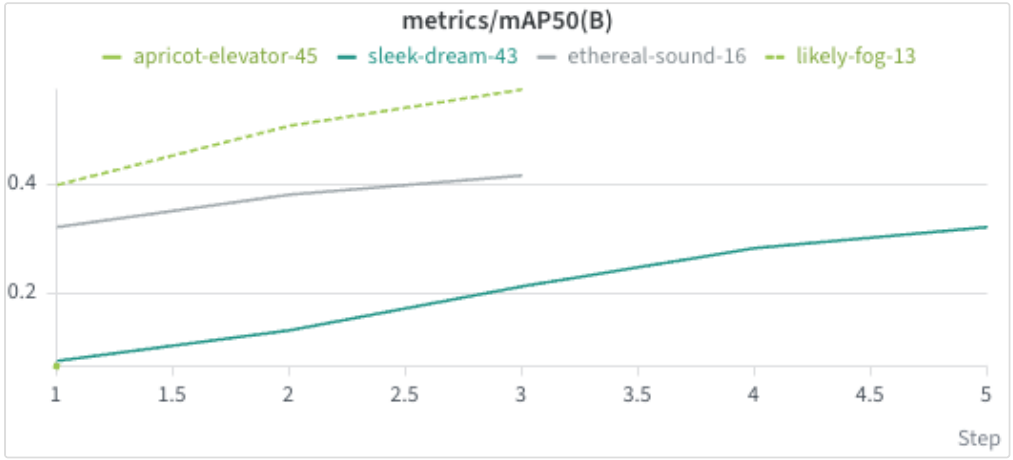



**Fig. 1** metrics/mAP50(B)

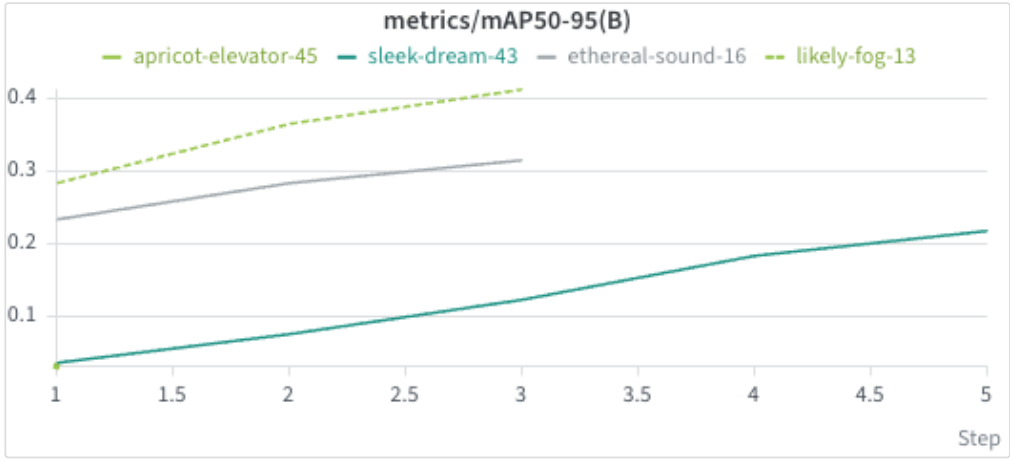



**Fig. 2** metrics/mAP50-95(B)

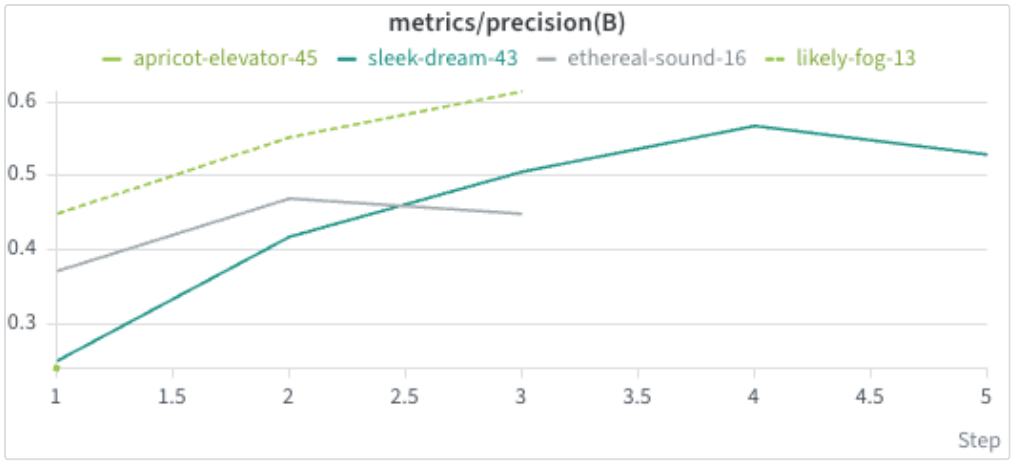



**Fig. 3** metrics/precision(B)

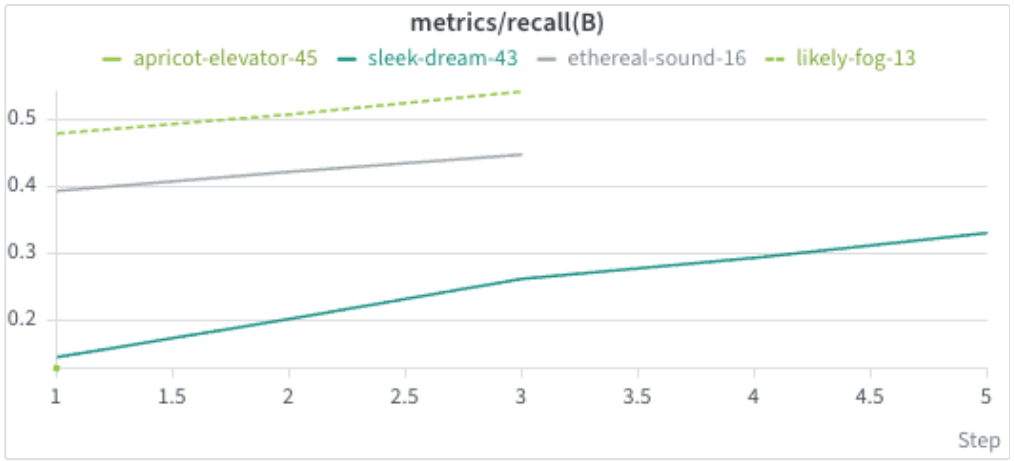



**Fig. 4** metrics/recall(B)

## 3 Training and validation losses

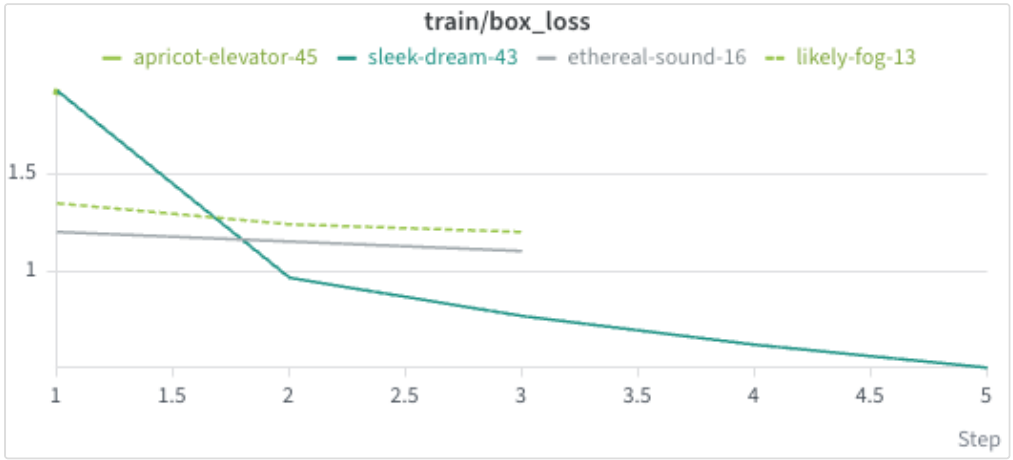



**Fig. 5** train/box_loss

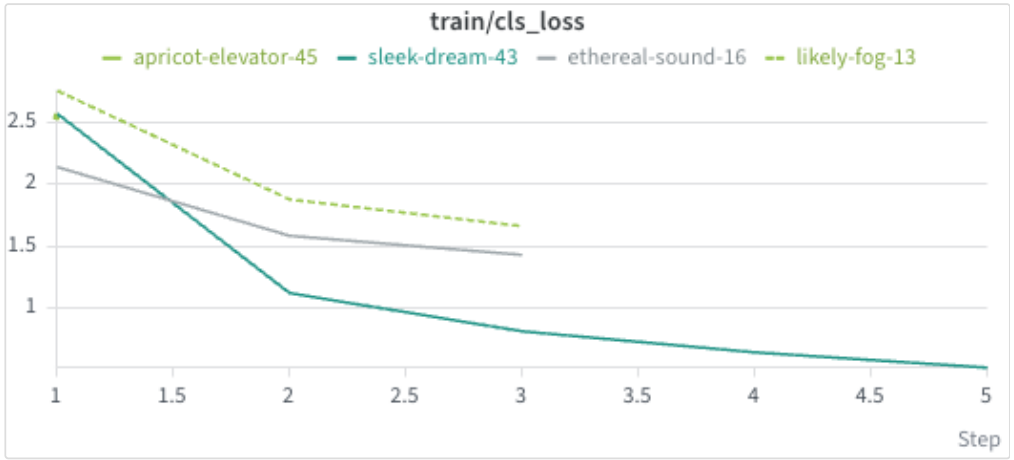



**Fig. 6** train/cls_loss

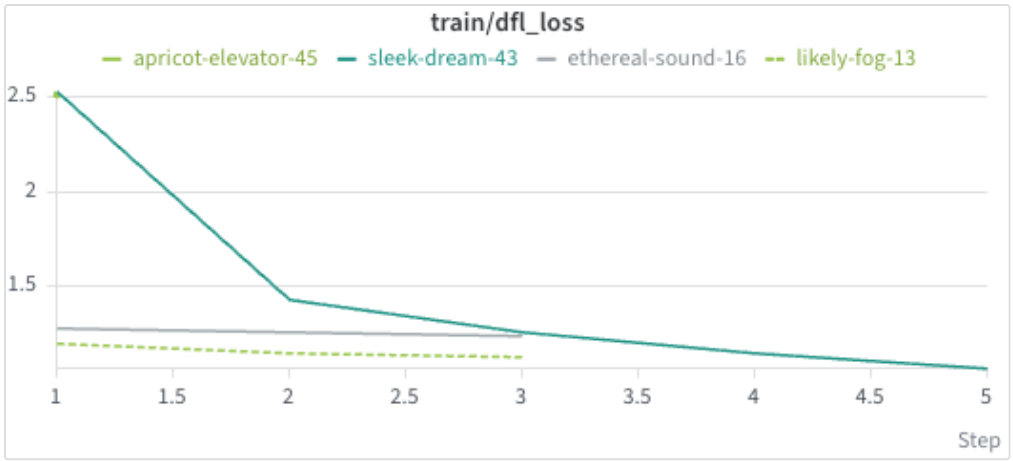



**Fig. 7** train/dfl_loss

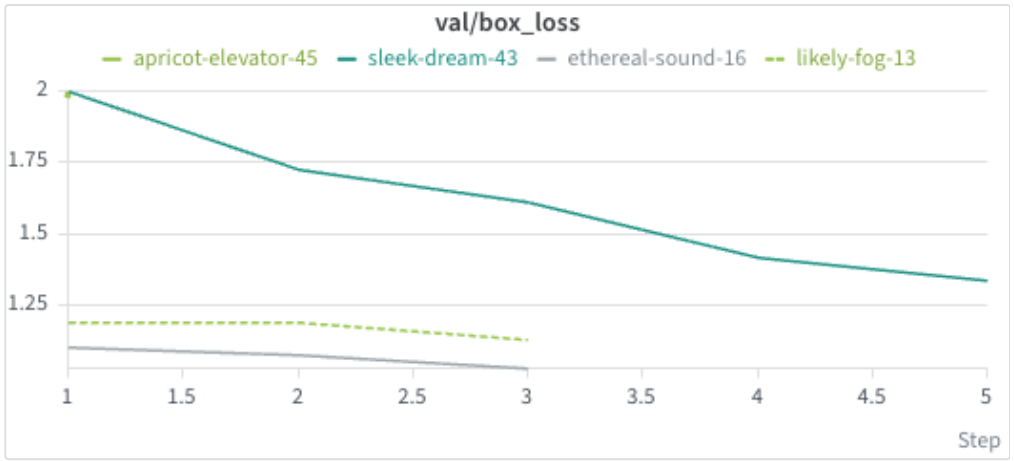



**Fig. 8** val/box_loss

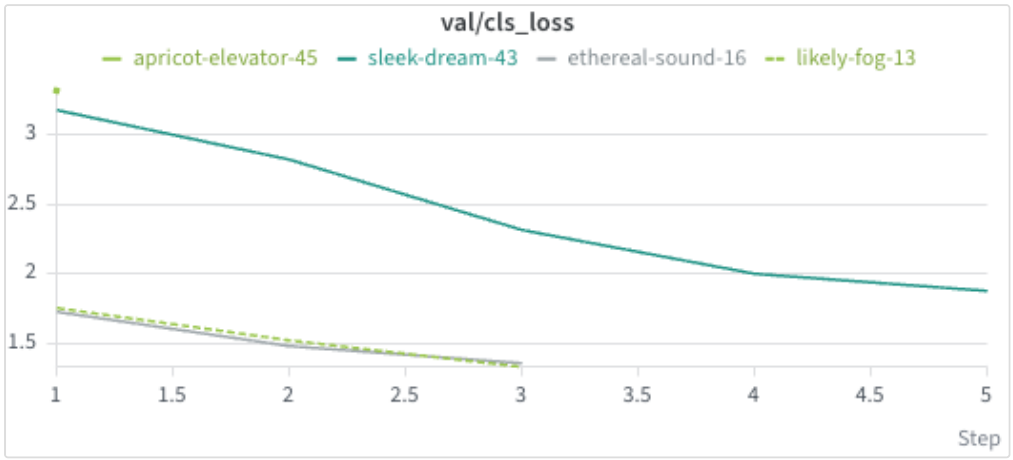



**Fig. 9** val/cls_loss

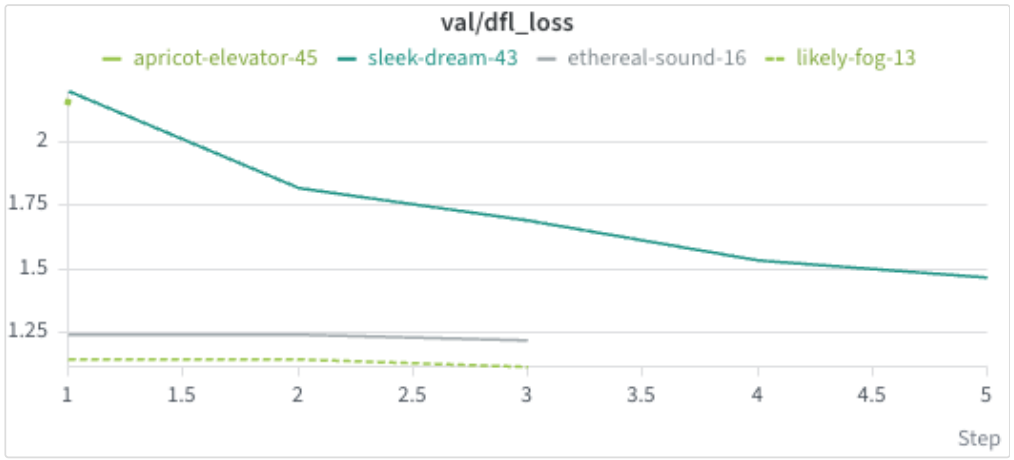



**Fig. 10** val/dfl_loss

## 4 Learning rate

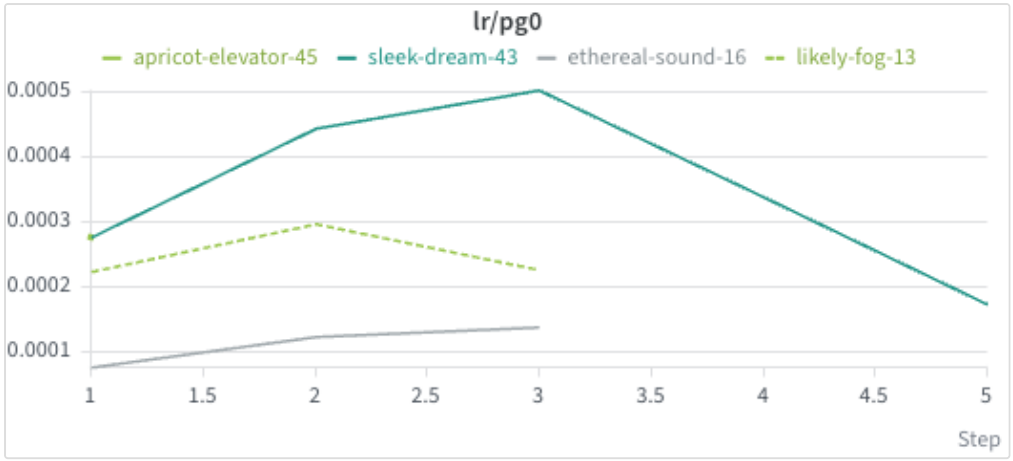



**Fig. 11** lr/pg0

## 5 Per-epoch values

| Run | Epoch | mAP50 | mAP50-95 | Precision | Recall | train box | train cls | train dfl | val box | val cls | val dfl | lr pg0 | Elapsed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| likely-fog-13 | 1 | 0.4021 | 0.2855 | 0.4496 | 0.4806 | 1.3513 | 2.7678 | 1.1894 | 1.1901 | 1.7639 | 1.1485 | 0.00022 | 2 min |
| | 2 | 0.5116 | 0.3658 | 0.5531 | 0.5095 | 1.2486 | 1.8802 | 1.1403 | 1.1857 | 1.5286 | 1.1482 | 0.00030 | 3 min |
| | 3 | 0.5777 | 0.4127 | 0.6142 | 0.5424 | 1.2062 | 1.6612 | 1.1201 | 1.1299 | 1.3378 | 1.1179 | 0.00023 | 3 min |
| ethereal-sound-16 | 1 | 0.3234 | 0.2336 | 0.3715 | 0.3938 | 1.2040 | 2.1399 | 1.2722 | 1.0987 | 1.7307 | 1.2425 | 0.00008 | 54 min |
| | 2 | 0.3837 | 0.2837 | 0.4706 | 0.4238 | 1.1541 | 1.5805 | 1.2551 | 1.0749 | 1.4885 | 1.2471 | 0.00012 | 81 min |
| | 3 | 0.4181 | 0.3149 | 0.4491 | 0.4473 | 1.1144 | 1.4312 | 1.2362 | 1.0271 | 1.3631 | 1.2225 | 0.00014 | 108 min |
| sleek-dream-43 | 1 | 0.0743 | 0.0374 | 0.2492 | 0.1453 | 1.9264 | 2.5775 | 2.5341 | 2.0002 | 3.1757 | 2.2003 | 0.00028 | 17 min |
| | 2 | 0.1294 | 0.0764 | 0.4191 | 0.2025 | 0.9698 | 1.1282 | 1.4307 | 1.7234 | 2.8228 | 1.8223 | 0.00045 | 26 min |
| | 3 | 0.2114 | 0.1242 | 0.5067 | 0.2625 | 0.7803 | 0.8137 | 1.2544 | 1.6093 | 2.3161 | 1.6918 | 0.00050 | 34 min |
| | 4 | 0.2842 | 0.1850 | 0.5681 | 0.2933 | 0.6373 | 0.6375 | 1.1435 | 1.4178 | 2.0029 | 1.5335 | 0.00034 | 42 min |
| | 5 | 0.3238 | 0.2180 | 0.5285 | 0.3303 | 0.5174 | 0.5166 | 1.0607 | 1.3385 | 1.8782 | 1.4719 | 0.00017 | 44 min |
| apricot-elevator-45 | 1 | 0.0635 | 0.0326 | 0.2397 | 0.1286 | 1.9166 | 2.5581 | 2.5184 | 1.9879 | 3.3158 | 2.1526 | 0.00028 | 12 min |

## 6 Run sleek-dream-43 25.3 M parameters · 8 classes · 5 epochs · mAP50 0.324



**Fig. 12** sleek-dream-43 · val_batch0_labels.jpg

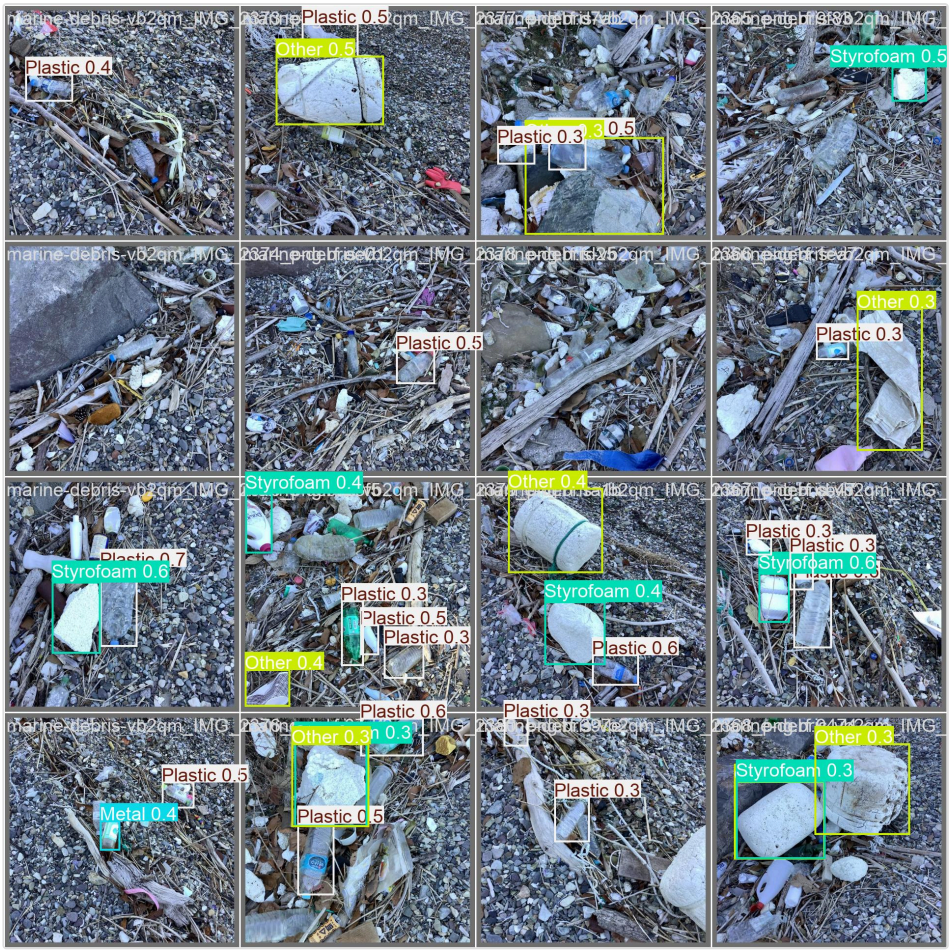


**Fig. 13** sleek-dream-43 · val_batch0_pred.jpg

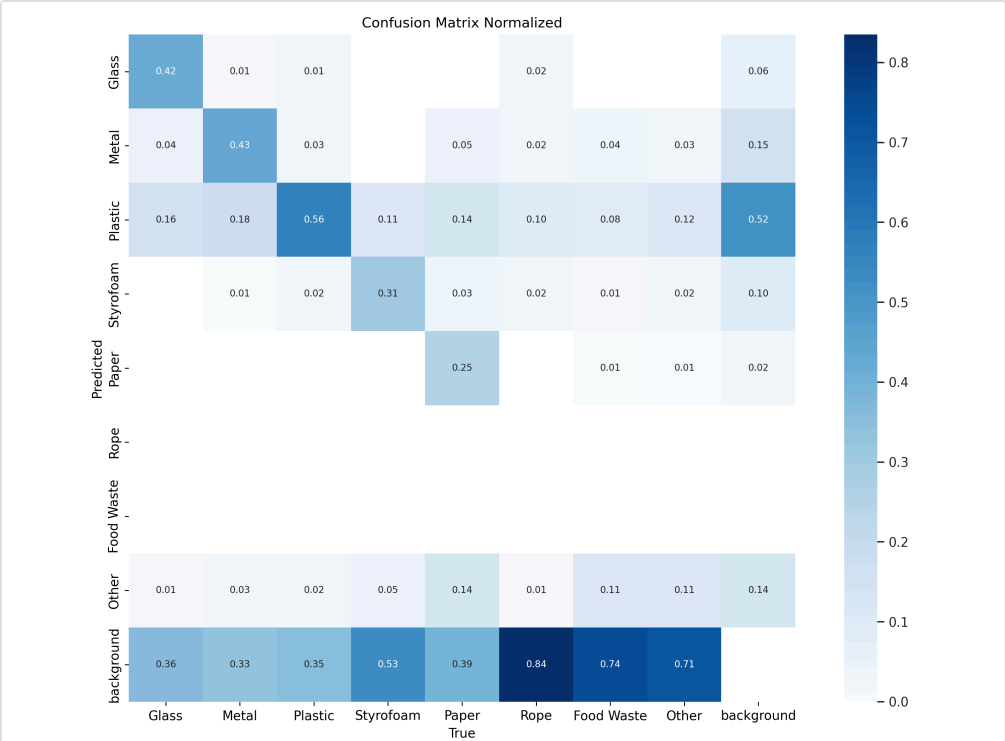



**Fig. 14** sleek-dream-43 · confusion_matrix_normalized.png

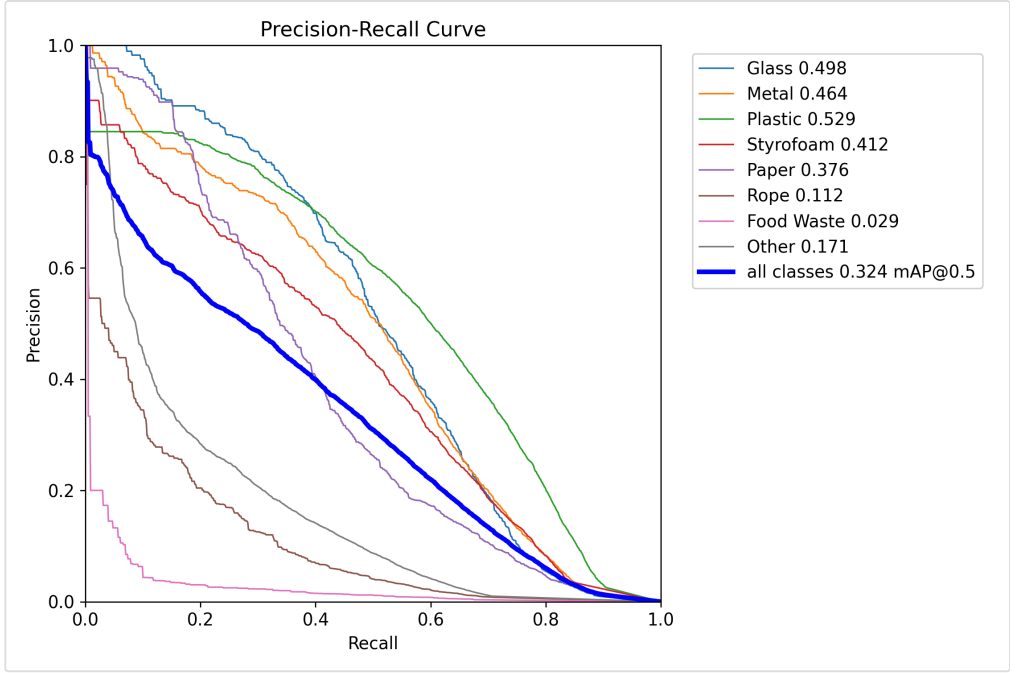



**Fig. 15** sleek-dream-43 · PR_curve.png

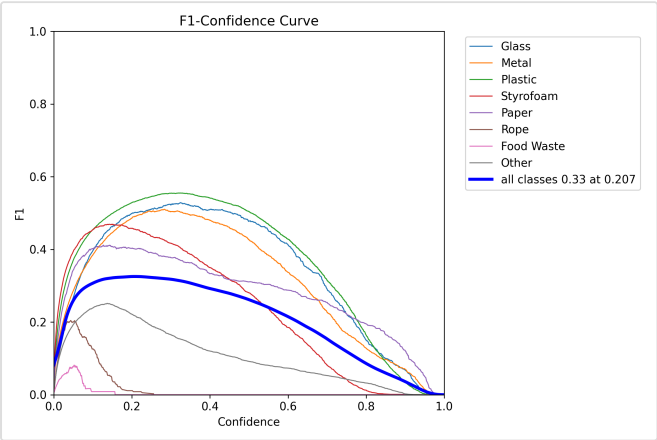



**Fig. 16** sleek-dream-43 · F1_curve.png

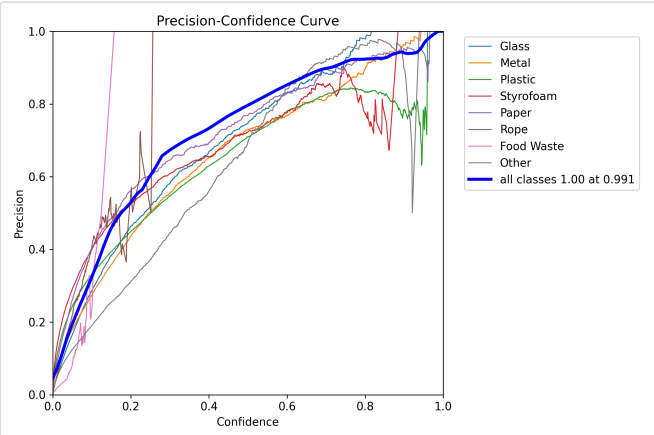



**Fig. 17** sleek-dream-43 · P_curve.png

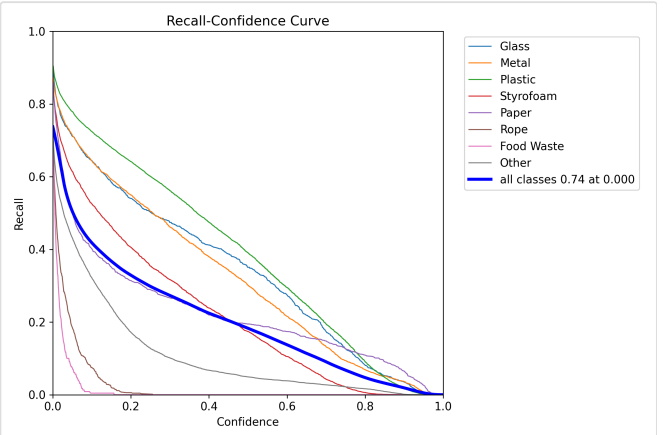



**Fig. 18** sleek-dream-43 · R_curve.png

## 7 Run likely-fog-13 3.0 M parameters · 12 classes · 3 epochs · mAP50 0.578

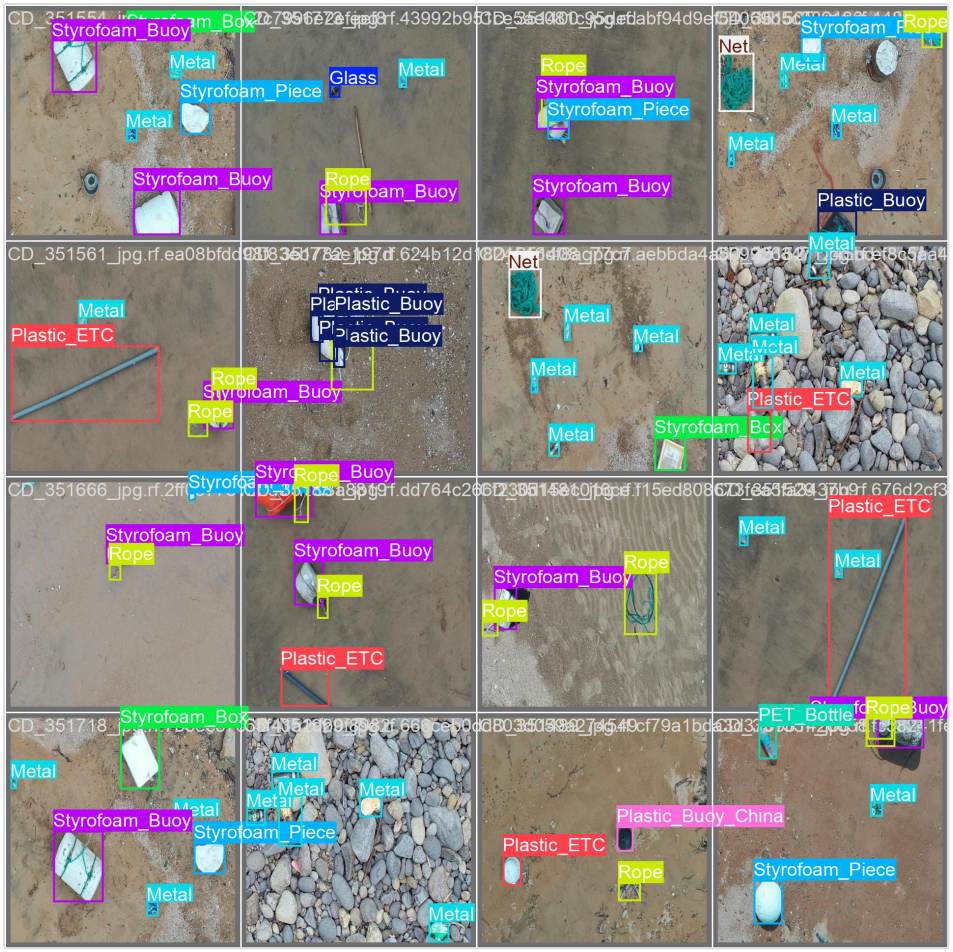


**Fig. 19** likely-fog-13 · val_batch0_labels.jpg

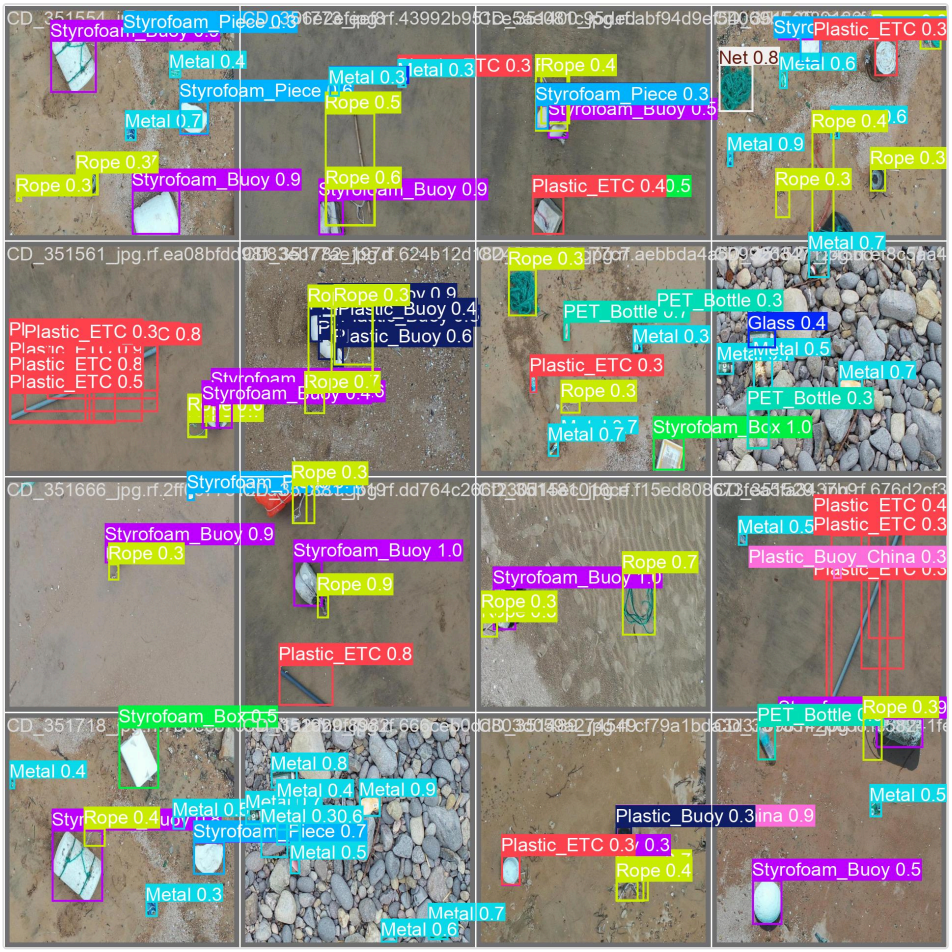


**Fig. 20** likely-fog-13 · val_batch0_pred.jpg

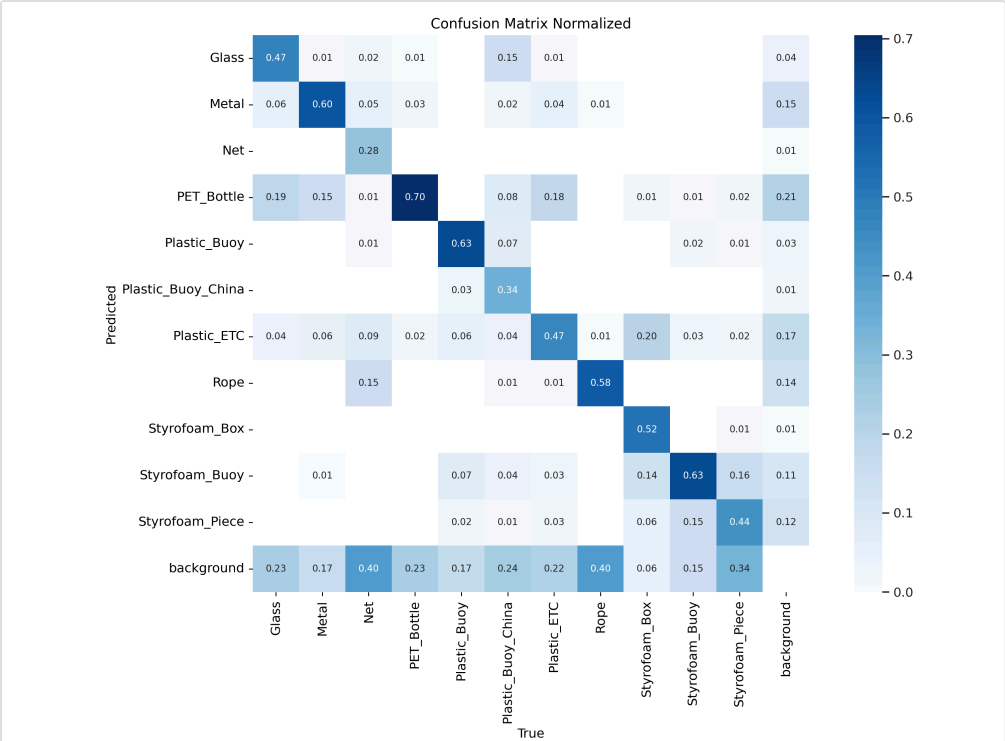



**Fig. 21** likely-fog-13 · confusion_matrix_normalized.png

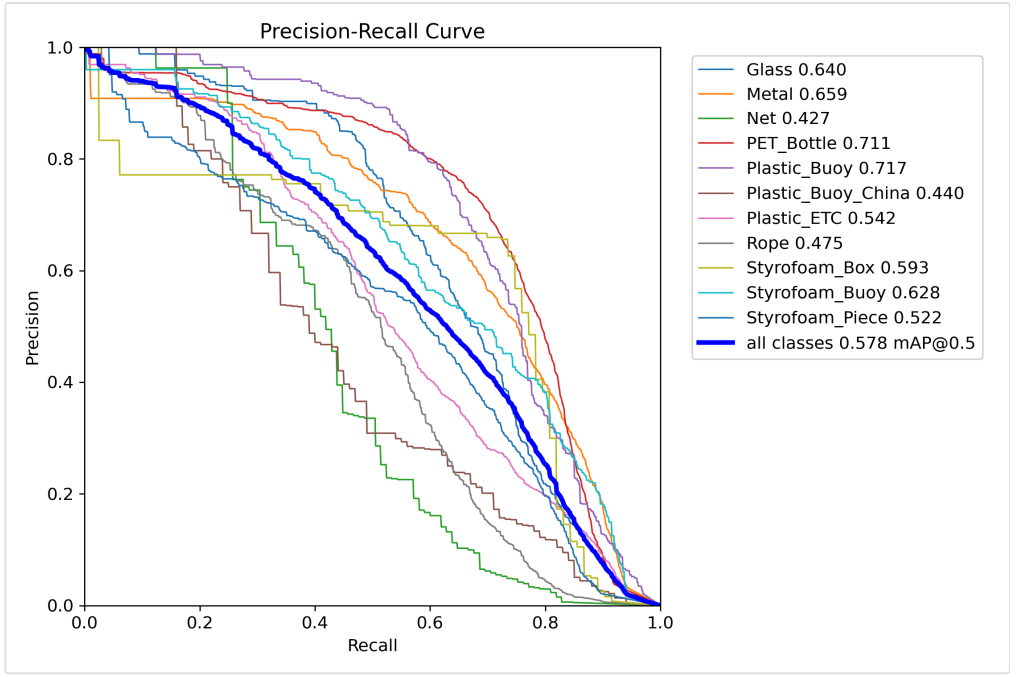



**Fig. 22** likely-fog-13 · PR_curve.png

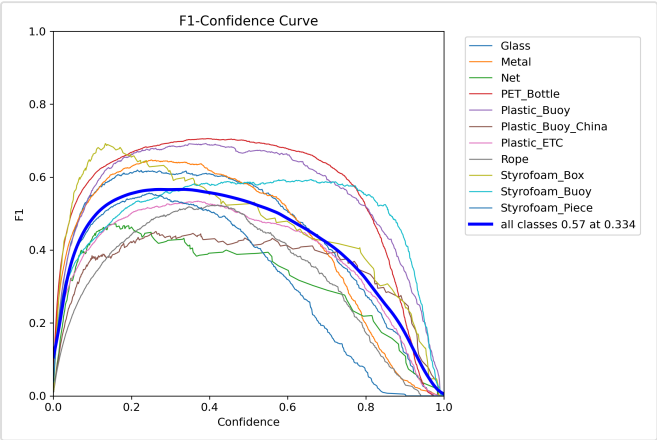



**Fig. 23** likely-fog-13 · F1_curve.png

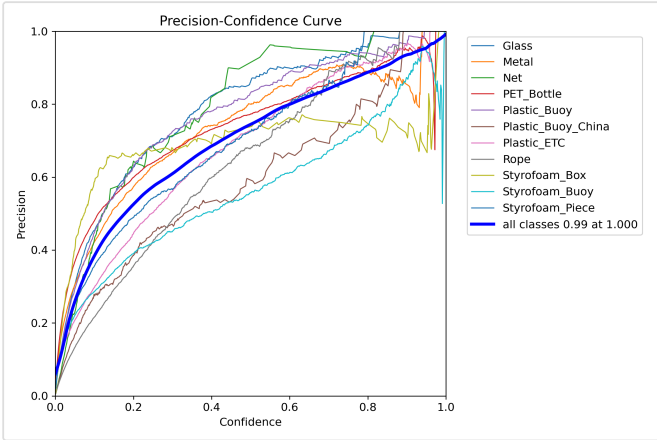



**Fig. 24** likely-fog-13 · P_curve.png

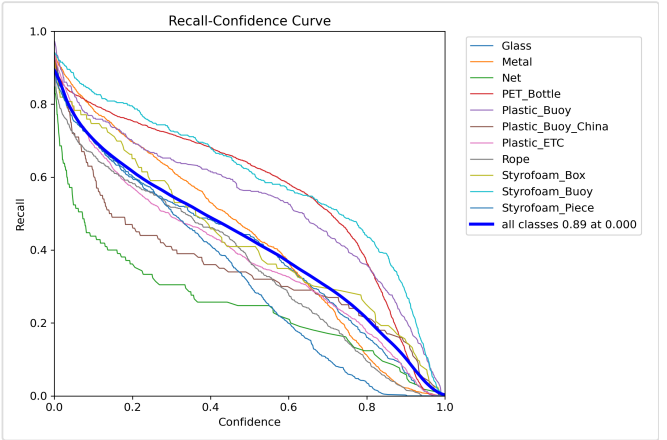



**Fig. 25** likely-fog-13 · R_curve.png

# 8 Dataset statistics

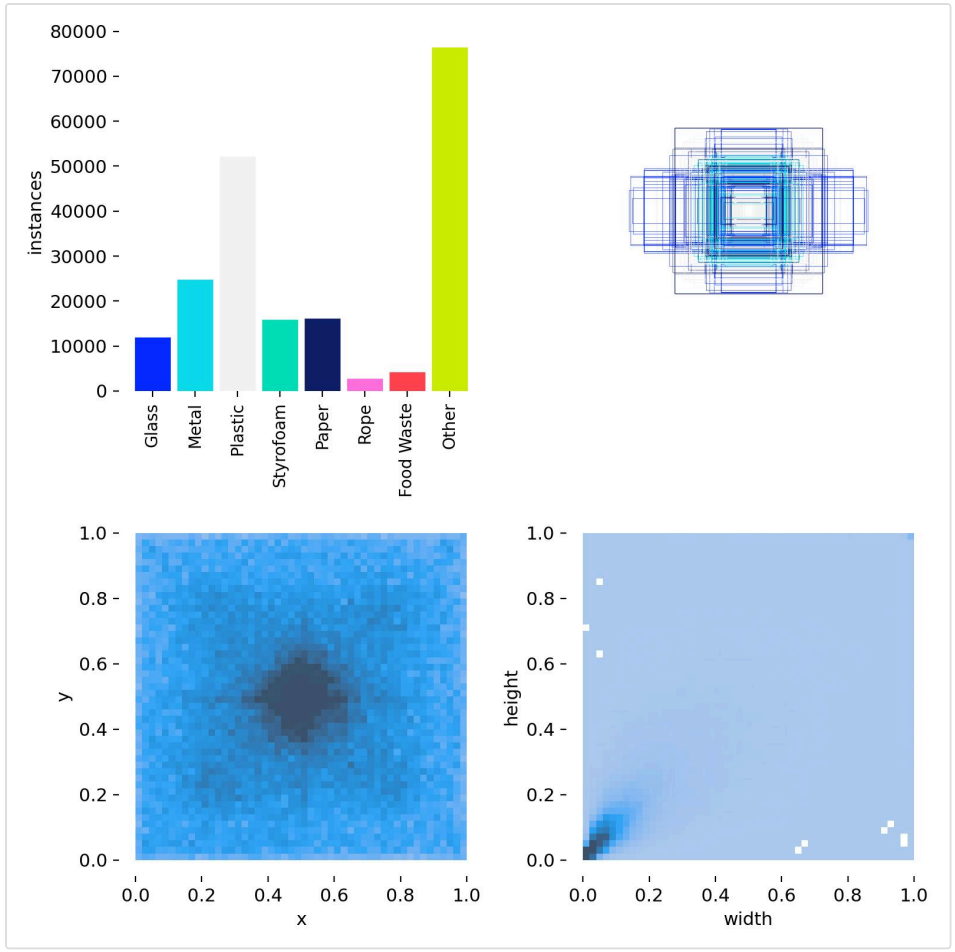



**Fig. 26** sleek-dream-43 · labels.jpg

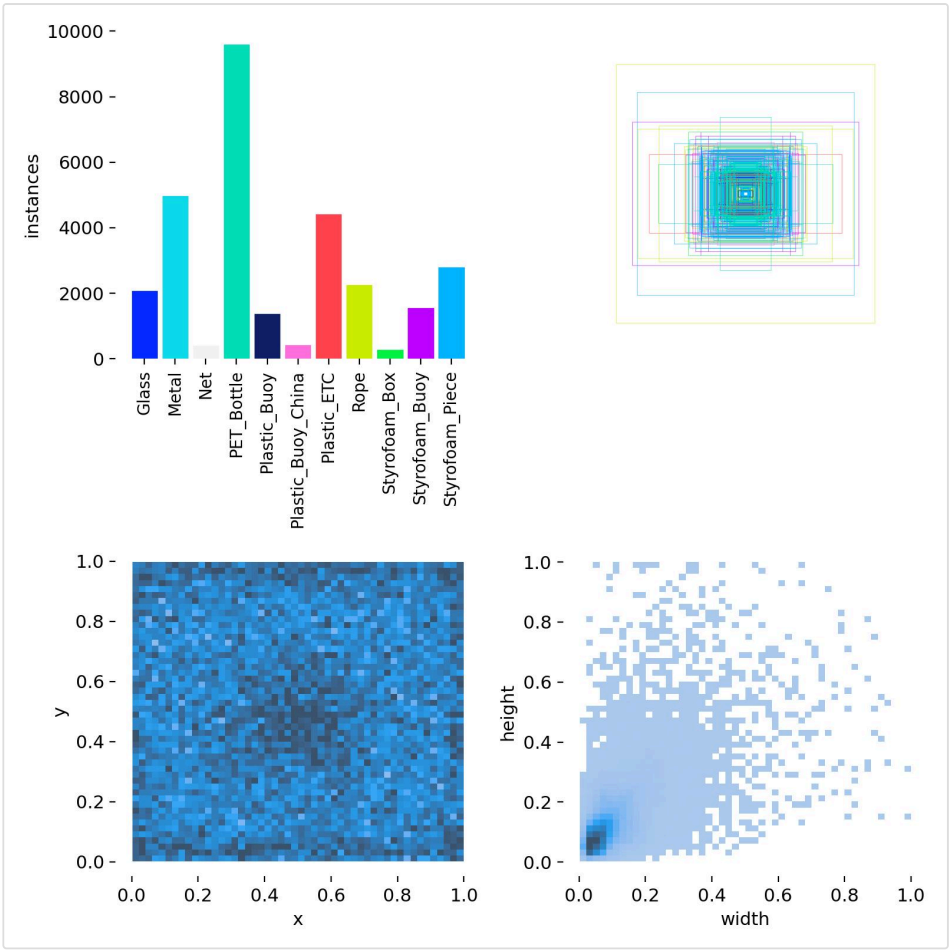



**Fig. 27** likely-fog-13 · labels.jpg

# A Appendix: system metrics

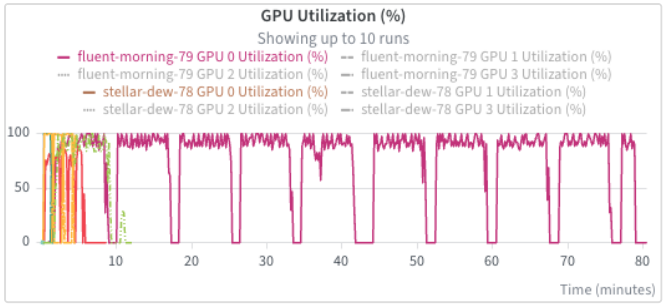



**Fig. 28** GPU utilization (%)

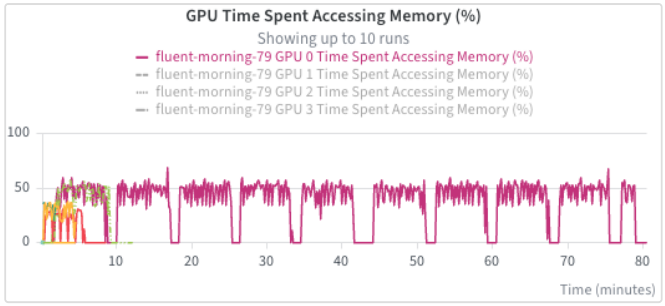



**Fig. 29** GPU time spent accessing memory (%)

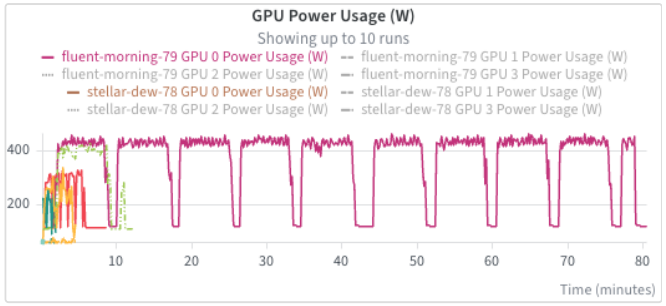



**Fig. 30** GPU power usage (W)

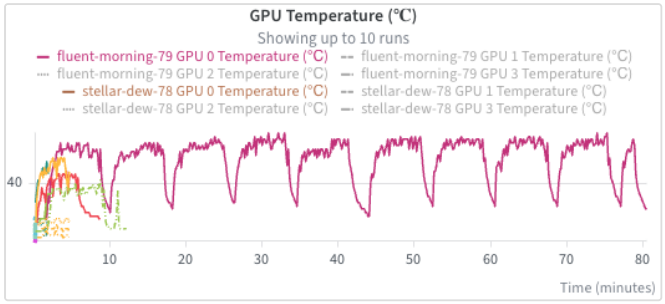



**Fig. 31** GPU temperature (°C)

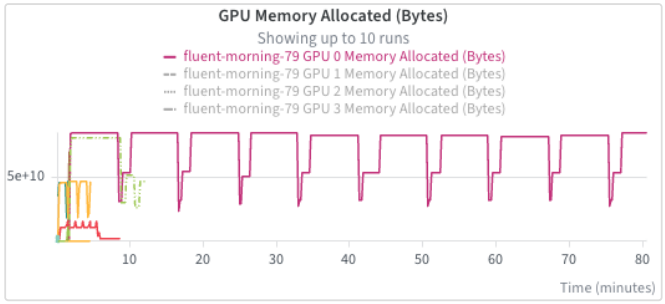



**Fig. 32** GPU memory allocated (bytes)

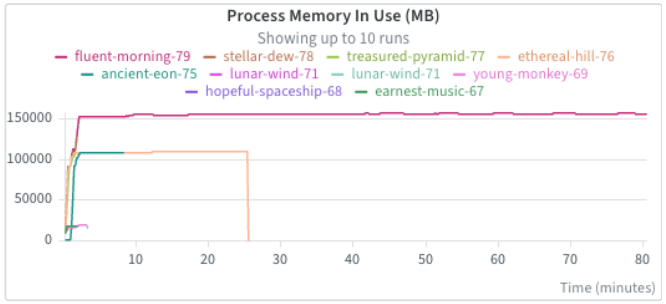



**Fig. 33** Process memory in use (MB)

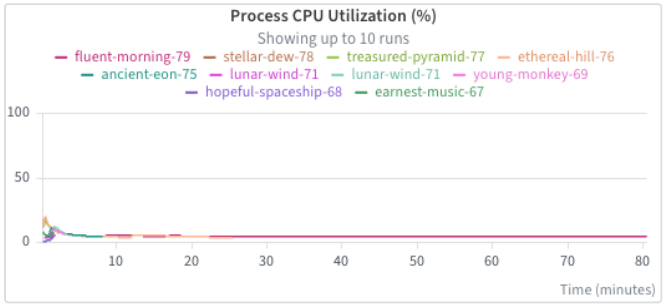



**Fig. 34** Process CPU utilization (%)

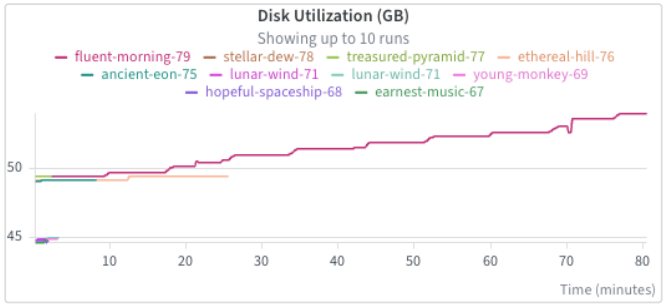



**Fig. 35** Disk utilization (GB)

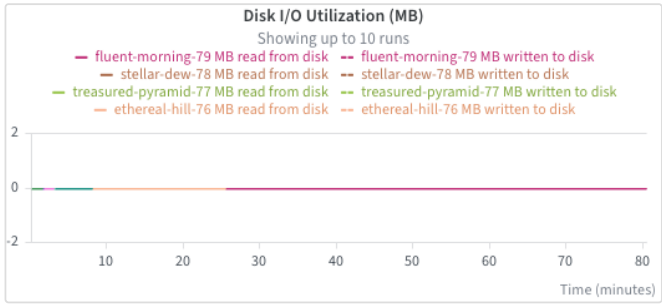



**Fig. 36** Disk I/O utilization (MB)

## B Appendix: run inventory

| Date | finished | crashed | failed | killed | Runs | With metrics |
|---|---|---|---|---|---|---|
| 2024-10-16 | 4 | 1 | · | · | 5 | · |
| 2024-10-21 | 9 | 4 | 1 | 6 | 20 | 2 |
| 2024-10-23 | 7 | 3 | 10 | · | 20 | 2 |
| 2024-11-06 | 5 | 7 | 20 | 2 | 34 | · |
| **All** | **25** | **15** | **31** | **8** | **79** | **4** |